Height bound

Every search halves the remaining keys, so after $O(\log n)$ levels we reach the bottom.

Theorem

The height of a random tower over n keys is $O(\log n)$ with high probability.

Proof.

- Let $H(y)$ be the tower height of key y, and $H = \max H(y)$.
- A key reaches level k only after k heads in a row, so $\Pr(H(y) \geq k) \leq 2^{-k}$.
- The tower is tall when one key is tall:
  $\Pr(H \geq k) = \Pr(H(1) \geq k \vee H(2) \geq k \vee \cdots \vee H(n) \geq k)$
- By the union bound – $\Pr(A \vee B) \leq \Pr(A) + \Pr(B)$ for any events A and B – we get:
  $\Pr(H \geq k) \leq \sum_{y=1}^{n} \Pr(H(y) \geq k) = n \cdot \Pr(H(y) \geq k) = \frac{n}{2^k}$
- For any $d > 2$, with $k = d \cdot \log_2 n$ we have $\Pr(H \geq d \log_2 n) \leq \frac{n}{2^{d \cdot \log_2 n}} = \frac{n}{n^d} = \frac{1}{n^{d-1}} < \frac{1}{n}$

□

Treaps

A treap keeps keys in search order and priorities in heap order.